

Distributional Reinforcement Learning | Part 2

Jonas Müller

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Outline

- 1 Recap
- 2 Linear Approximation
- 3 Theory when allowing signed measures
- 4 Distributional DQN
- 5 Particle DQN

Recap: MDP and Return

We are working on a finite MDP:

$$(\mathcal{X}, \mathcal{A}, \xi_0, P_X, P_R), \quad \pi : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{A}), \quad \gamma \in [0, 1).$$

Trajectory process:

$$\begin{aligned} X_0 &\sim \xi_0, & A_t &\sim \pi(\cdot \mid X_t), \\ R_t &\sim P_R(\cdot \mid X_t, A_t), & X_{t+1} &\sim P_X(\cdot \mid X_t, A_t), \end{aligned} \quad t \geq 0.$$

Discounted return:

$$G^\pi(x) = \sum_{t=0}^{\infty} \gamma^t R_t \quad \text{given } X_0 = x.$$

Recap: Value function and Bellman Operator

The value function is given by the expected return:

$$V^\pi(x) = \mathbb{E}[G^\pi(x)] = \int_{\mathbb{R}} z \, d\eta_\pi(x)(z)$$

Where $\eta_\pi(x) = \mathcal{D}(G^\pi(x))$ is the return distribution.

As we know, there is a recursive relation for the value function known as Bellman's Equation:

$$V^\pi = \mathcal{T}^\pi V^\pi.$$

Where we have the Bellman operator $\mathcal{T}^\pi : \mathbb{R}^{\mathcal{X}} \rightarrow \mathbb{R}^{\mathcal{X}}$:

$$(\mathcal{T}^\pi V)(x) = \mathbb{E}_\pi [R + \gamma V(X') \mid X = x].$$

Recap: Random-Variable Bellman Equation

With (A, R, X') sampled from $\pi(\cdot | x)P_R(\cdot | x, A)P_X(\cdot | x, A)$, the random-variable Bellman equation is:

$$G^\pi(x) \stackrel{D}{=} R + \gamma G^\pi(X') \mid X = x.$$

Taking laws gives the return-distribution equation:

$$\eta_\pi(x) = \mathcal{D}(R + \gamma G^\pi(X') \mid X = x).$$

This is the distributional version of $(\mathcal{T}^\pi V)(x) = \mathbb{E}_\pi[R + \gamma V(X') \mid X = x]$.

Recap: Distributional Bellman Operator

Define the bootstrap map

$$b_{r,\gamma}(z) = r + \gamma z, \quad r, z \in \mathbb{R},$$

and for $Z' \sim \eta(x')$,

$$(b_{r,\gamma})_{\#}\eta(x') = \mathcal{D}(r + \gamma Z').$$

Distributional Bellman operator $\mathcal{T}^{\pi} : \mathcal{P}(\mathbb{R})^{\mathcal{X}} \rightarrow \mathcal{P}(\mathbb{R})^{\mathcal{X}}$:

$$(\mathcal{T}^{\pi}\eta)(x) = \mathbb{E}_{\pi} [(b_{R,\gamma})_{\#}\eta(X') \mid X = x].$$

Distributional Bellman equation:

$$\eta_{\pi}(x) = \mathbb{E}_{\pi} [(b_{R,\gamma})_{\#}\eta_{\pi}(X') \mid X = x], \quad \eta_{\pi} = \mathcal{T}^{\pi}\eta_{\pi}.$$

Recap: Representations of Return Distributions

Categorical representation:

$$\eta(x) = \sum_{i=1}^m p_i^\theta(x) \delta_{\theta_i}, \quad p_i^\theta(x) \geq 0, \quad \sum_{i=1}^m p_i^\theta(x) = 1.$$

Quantile representation:

$$\eta(x) = \frac{1}{m} \sum_{i=1}^m \delta_{\theta_i(x)}, \quad \tau_i = \frac{2i-1}{2m}, \quad \theta_i(x) \approx F_{\eta(x)}^{-1}(\tau_i).$$

Focus of this Presentation

If $\mathcal{X}$ is large, then storing even a scalar, let alone a distribution per state is expensive. Hence, we look for suitable approximations.

Linear Value Function Approximation | Scalar Case

For weights $w \in \mathbb{R}^n$ approximate the value function as

$$V_w(x) = \phi(x)^\top w, \quad \nabla_w V_w(x) = \phi(x).$$

For finite $\mathcal{X}$ and feature map ϕ , stack feature rows into $\Phi \in \mathbb{R}^{|\mathcal{X}| \times n}$:

$$V_w = \Phi w, \quad \mathcal{F}_\phi = \{\Phi w : w \in \mathbb{R}^n\}.$$

For state-action pairs, use e.g. $\phi : \mathcal{X} \times \mathcal{A} \rightarrow \mathbb{R}^n$ via $Q_w(x, a) = \phi(x, a)^\top w$.

Idea

Instead of having a lookup table for the value of each state, we use a linear function to represent the function. **This function can be queried cheaply.**

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Question: How do we get a good w ?

Linear Approximation as Regression

Fitting a linear value function is just ordinary least squares:

$$V_w = \Phi w, \quad \min_{w \in \mathbb{R}^n} \|V^\pi - V_w\|_2.$$

which we could solve by the normal equation.

However

In reinforcement learning, we might want to have a better approximation for some states than others.

Weighted Norm for Linear Approximation

For some $\xi \in \mathcal{P}(\mathcal{X})$ we define the weighted L_2 norm and diagonal weight matrix

$$\|V - V'\|_{\xi,2} = \left(\sum_{x \in \mathcal{X}} \xi(x) (V(x) - V'(x))^2 \right)^{1/2}. \quad \Xi = \text{diag}(\xi(x) : x \in \mathcal{X}).$$

The problem we now want to solve is:

$$\min_{w \in \mathbb{R}^n} \|V^\pi - V_w\|_{\xi,2}.$$

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Proposition 9.3 | Weighted Normal Equation

If ξ has full support and the columns of Φ are linearly independent, the unique solution is

$$w^* = (\Phi^\top \Xi \Phi)^{-1} \Phi^\top \Xi V^\pi.$$

Projection Back into the Feature Space

We have a linear approximation now, however we can't iterate the Bellman operator because we could have

$$\mathcal{T}^\pi V_w \notin \mathcal{F}_\phi$$

Define the weighted orthogonal projection

$$(\Pi_{\phi,\xi} V)(x) = \phi(x)^\top w^*, \quad w^* \in \arg \min_w \|V - V_w\|_{\xi,2}.$$

Approximate dynamic programming therefore iterates

$$V_{k+1} = \Pi_{\phi,\xi} \mathcal{T}^\pi V_k.$$

- $\mathcal{T}^\pi$ performs the Bellman update.
- $\Pi_{\phi,\xi}$ removes components that the features cannot express.

Repeated projection can compound error (the diffusion effect).

When Is the Projected Operator Stable?

Let ξ_π be a fully supported invariant distribution under π .

Projected Bellman contraction (Bellemare et al., 2023, Theorem 9.8)

Since Π_{ϕ, ξ_π} is nonexpansions in $\|\cdot\|_{\xi_\pi, 2}$,

$$\|\Pi_{\phi, \xi_\pi} \mathcal{T}^\pi V - \Pi_{\phi, \xi_\pi} \mathcal{T}^\pi V'\|_{\xi_\pi, 2} \leq \gamma \|V - V'\|_{\xi_\pi, 2}.$$

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How good is this compared to the true best value function?

$\hat{V}^\pi = \Pi_{\phi, \xi_\pi} \mathcal{T}^\pi \hat{V}^\pi$, with

$$\|\hat{V}^\pi - V^\pi\|_{\xi_\pi, 2} \leq \frac{1}{\sqrt{1 - \gamma^2}} \|\Pi_{\phi, \xi_\pi} V^\pi - V^\pi\|_{\xi_\pi, 2}.$$

From Value function approximation to Distributional Approximation

Want to linearly approximate probability distributions. However in general

$$\eta_{w_1}(x) + \eta_{w_2}(x) \notin \mathcal{P}(\mathbb{R})$$

and since the distribution lives in $\mathcal{P}(\mathbb{R})$, we can have non finite support.

How to solve this?

Fix a finite dimensional family of distributions $\mathcal{F}_\theta \subset \mathcal{P}(\mathbb{R})$ and then linearly approximate the parameters of the distribution. “Linear” distributional approximation means that the *parameters of a probability representation* depend linearly on features ϕ .

There are therefore two approximation layers:

- 1 a finite probability representation (categorical or quantile),
- 2 linear value function approximation.

Finite Return-Distribution Families

For a probability measure $\eta \in \mathcal{P}(\mathbb{R})$, we have

$$F_\eta(t) = \eta((-\infty, t]), \quad F_\eta^{-1}(\tau) = \inf\{z \in \mathbb{R} : F_\eta(z) \geq \tau\}.$$

We consider the finite dimensional approximations:

$$\text{categorical:} \quad \eta(x) = \sum_{i=1}^m p_i(x) \delta_{\theta_i}, \quad p_i(x) \geq 0, \quad \sum_{i=1}^m p_i(x) = 1,$$

$$\text{quantile:} \quad \eta(x) = \frac{1}{m} \sum_{i=1}^m \delta_{\theta_i(x)}, \quad \tau_i = \frac{2i-1}{2m}, \quad \theta_i(x) \approx F_{\eta(x)}^{-1}(\tau_i).$$

What is state-dependent?

Categorical representations fix θ_i and learn the probabilities $p_i(x)$. Quantile fixes equal probabilities $1/m$ and learns locations $\theta_i(x) \approx F_{\eta(x)}^{-1}(\tau_i)$.

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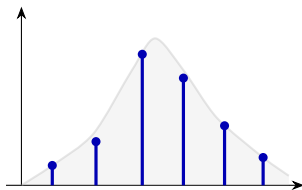
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Question: We do not know $F_{\eta(x)}^{-1}$ in practice.

Categorical vs. Quantile

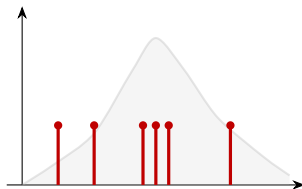
Categorical

$$\eta(x) = \sum_{i=1}^m p_i(x) \delta_{\theta_i}$$

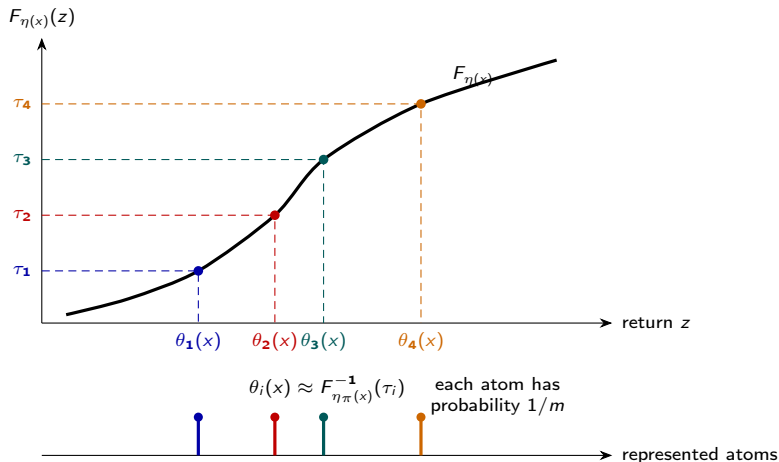


Quantile

$$\eta(x) = \frac{1}{m} \sum_{i=1}^m \delta_{\theta_i(x)}$$



Quantile Functions



Distributional TD: Linear Learning

For fixed feature map $\phi(x) \in \mathbb{R}^d$. A linear method has, for every atom,

$$u_i(x) = \phi(x)^\top w_i, \quad w_i \in \mathbb{R}^d.$$

Observing a transition (x, r, x') gives us an empirical target. We fit this via

$$w_i \leftarrow w_i + \alpha \varepsilon_i(x, r, x') \phi(x).$$

where ε_i is an error term. This term and the meaning of u_i is what changes.

Linear Quantile TD: Representation and Loss

For the quantile representation, we choose

$$u_i(x) = \theta_i(x) = \phi(x)^\top w_i,$$

The represented return distribution is

$$\eta_w(x) = \frac{1}{m} \sum_{i=1}^m \delta_{\theta_i(x)} = \frac{1}{m} \sum_{i=1}^m \delta_{\phi(x)^\top w_i},$$

Then we take the quantile loss as in the tabular case

$$\rho_\tau(\Delta) = |\mathbb{1}_{\{\Delta < 0\}} - \tau| |\Delta|.$$

For one target z , atom i uses this loss on the residual

$$L_{\tau_i}(w_i) = \rho_{\tau_i}(z - \phi(x)^\top w_i),$$

whose subgradient w.r.t. w_i is $-(\tau_i - \mathbb{1}_{\{z < \phi(x)^\top w_i\}})\phi(x)$.

Linear QTD: Distributional Semi-Gradient

Similar to the tabular case, for transition (x, a, r, x') we get m target samples

$$g_j = r + \gamma \phi(x')^\top w_j, \quad j = 1, \dots, m.$$

For atom i , average the quantile loss over these targets:

$$L_i(w_i) = \frac{1}{m} \sum_{j=1}^m \rho_{\tau_i}(g_j - \phi(x)^\top w_i).$$

The semi-gradient step is

$$w_i \leftarrow w_i - \alpha \nabla_{w_i} L_i(w_i) = w_i + \alpha \frac{1}{m} \sum_{j=1}^m \underbrace{(\tau_i - \mathbb{1}_{\{g_j < \phi(x)^\top w_i\}})}_{\text{quantile TD error}} \phi(x).$$

Unlike the scalar TD error, each averaged quantile error is in $[-1, 1]$.

Linear Categorical TD: Prediction

We approximate the probabilities $p_i(x; w)$ by a linear combination of features:

$$p_i(x; w) = \frac{\exp(\phi(x)^\top w_i)}{\sum_{j=1}^m \exp(\phi(x)^\top w_j)}, \quad \eta_w(x) = \sum_{i=1}^m p_i(x; w) \delta_{\theta_i}.$$

transformed into probabilities by the softmax function. Softmax is needed, direct linear coefficients need not be nonnegative or sum to one.

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This can be handled, later!

Linear CTD: How to Stay on Fixed Support

For a sampled transition (x, r, x') , the Bellman update is an update of the prediction at the current state x :

$$\underbrace{\eta_w(x) = \sum_{j=1}^m p_j(x; w) \delta_{\theta_j}}_{\text{prediction at } x} \xrightarrow{\text{Observed } (r, x')} \underbrace{(b_{r, \gamma})_{\#} \eta_w(x') = \sum_{j=1}^m p_j(x'; w) \delta_{r + \gamma \theta_j}}_{\text{Empirical Bellman step for } x}.$$

Since in general $r + \gamma \theta_j$ need not lie on the fixed support, use the categorical projection Π_c to project the shifted atoms back to $S = \{\theta_1, \dots, \theta_m\}$. For each j , if $\theta_\ell \leq r + \gamma \theta_j \leq \theta_{\ell+1}$, split its mass as

$$\bar{p}_\ell += p_j(x'; w) \frac{\theta_{\ell+1} - (r + \gamma \theta_j)}{\theta_{\ell+1} - \theta_\ell}, \quad \bar{p}_{\ell+1} += p_j(x'; w) \frac{(r + \gamma \theta_j) - \theta_\ell}{\theta_{\ell+1} - \theta_\ell}.$$

This gives the projected target

$$\bar{\eta}_{x, r, x'} = \Pi_c(b_{r, \gamma})_{\#} \eta_w(x') = \sum_{i=1}^m \bar{p}_i \delta_{\theta_i}.$$

Linear CTD: Target, Loss, and Update

To learn softmax probabilities, we use the Cross-entropy loss:

$$L(w) = - \sum_{i=1}^m \bar{p}_i \log p_i(x; w).$$

Since

$$\nabla_{w_i} L(w) = -(\bar{p}_i - p_i(x; w)) \phi(x)$$

gradient descent gives the semi-gradient CTD update:

$$w_i \leftarrow w_i + \alpha \underbrace{(\bar{p}_i - p_i(x; w))}_{\text{CTD error}} \phi(x).$$

Signed Categorical Family

We can make our lives to develop theory easier by allowing negative measures. The signed categorical family on $S = \{\theta_1, \dots, \theta_m\}$ is

$$\mathcal{F}_{S,m} = \left\{ \sum_{i=1}^m p_i \delta_{\theta_i} \mid p_i \in \mathbb{R}, \sum_{i=1}^m p_i = 1 \right\} \subseteq \mathcal{M}(\mathbb{R}).$$

Using this we can define the signed categorical return functions

$$\mathcal{F}_{S,m}^{\mathcal{X}} = \{\eta : \mathcal{X} \rightarrow \mathcal{F}_{S,m}\}.$$

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How to linearly approximate this?

Linear Approximation of signed measures

Linear approximation, subtracting the mean!

$$p_i(x) = \frac{1}{m} + \phi(x)^\top w_i - \frac{1}{m} \sum_{j=1}^m \phi(x)^\top w_j.$$

Then

$$\sum_{i=1}^m p_i(x) = 1, \quad p_i(x) \in \mathbb{R}.$$

The resulting family of distribution is denoted by $\mathcal{F}_{\phi, S, m}$.

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We want to perform bellman updates and ensure that the result is still in $\mathcal{F}_{\phi, S, m}$!

Why Two Projections?

Start with a $\eta \in \mathcal{F}_{\phi, S, m}$. After the Bellman update,

$$\mathcal{T}^\pi \eta$$

two things can go wrong.

- 1. Distribution representation.** For a state x , $(\mathcal{T}^\pi \eta)(x)$ need not be supported on $S = \{\theta_1, \dots, \theta_m\}$. Hence we apply the categorical projection Π_c from before.
- 2. Linear feature representation.** Even after Π_c , the map $x \mapsto (\Pi_c \mathcal{T}^\pi \eta)(x)$ need not lie in $\mathcal{F}_{\phi, S, m}$. Hence we need to define the feature projection by

$$\Pi_{\phi, \xi, d} \tilde{\eta} \in \arg \min_{\eta' \in \mathcal{F}_{\phi, S, m}} d(\tilde{\eta}, \eta').$$

In some suitable metric d . The final bellman update is then

$$\Pi_{\phi, \xi, d} \Pi_c \mathcal{T}^\pi \eta.$$

The Cramér Distance

We use the Cramér distance introduced prior, given by

$$\ell_2^2(\nu, \nu') = \int_{\mathbb{R}} (F_\nu(z) - F_{\nu'}(z))^2 dz$$

whenever this quantity is finite. Define

$$\mathcal{M}_{\ell_2,1}(\mathbb{R}) = \{\nu \in \mathcal{M}(\mathbb{R}) : \nu(\mathbb{R}) = 1, \ell_2(\nu, \delta_0) < \infty\}.$$

For return-distribution functions $\eta : \mathcal{X} \rightarrow \mathcal{M}_{\ell_2,1}(\mathbb{R})$ we can again define a weighted distance

$$\ell_{\xi,2}^2(\eta, \eta') = \sum_{x \in \mathcal{X}} \xi(x) \ell_2^2(\eta(x), \eta'(x)).$$

Bellman and Categorical Projections

Assume the MDP has stationary distribution ξ_π under policy π .

Lemma 9.14: Bellman Step

Under these assumptions, $\mathcal{T}^\pi : \mathcal{M}_{\ell_2,1}(\mathbb{R})^{\mathcal{X}} \rightarrow \mathcal{M}_{\ell_2,1}(\mathbb{R})^{\mathcal{X}}$ is a $\gamma^{1/2}$ -contraction with respect to $\ell_{\xi_\pi,2}$:

$$\ell_{\xi_\pi,2}(\mathcal{T}^\pi \eta, \mathcal{T}^\pi \eta') \leq \gamma^{1/2} \ell_{\xi_\pi,2}(\eta, \eta').$$

Lemma 9.15: Categorical Projection

The pointwise categorical projection operator

$$\Pi_c : \mathcal{M}_{\ell_2,1}(\mathbb{R})^{\mathcal{X}} \rightarrow \mathcal{F}_{S,m}^{\mathcal{X}}, \quad (\Pi_c \eta)(x) = \Pi_c(\eta(x))$$

is a nonexpansion with respect to $\ell_{\xi_\pi,2}$:

$$\ell_{\xi_\pi,2}(\Pi_c \eta, \Pi_c \eta') \leq \ell_{\xi_\pi,2}(\eta, \eta').$$

Feature Projection

The final projection is the analogue of the scalar linear projection, but applied to signed categorical distributions with the Cramér metric.

Lemma 9.16: Linear Feature Projection

The function approximation projection $\Pi_{\phi, \xi_\pi, \ell_2} : \mathcal{F}_{S, m}^{\mathcal{X}} \rightarrow \mathcal{F}_{\phi, S, m}$ is a nonexpansion with respect to $\ell_{\xi_\pi, 2}$: for any $\eta, \eta' \in \mathcal{F}_{S, m}^{\mathcal{X}}$,

$$\ell_{\xi_\pi, 2}(\Pi_{\phi, \xi_\pi, \ell_2} \eta, \Pi_{\phi, \xi_\pi, \ell_2} \eta') \leq \ell_{\xi_\pi, 2}(\eta, \eta').$$

Thus each piece is either a contraction or a nonexpansion:

$$\mathcal{T}^\pi \quad \text{contracts,} \quad \Pi_c, \Pi_{\psi, \xi_\pi, \ell_2} \quad \text{do not expand.}$$

Theorem: Doubly Projected Bellman

Composing the three preceding Lemma gives us the main result.

Theorem 9.13

Under the assumptions above, for all $\eta, \eta' \in \mathcal{M}_{\ell_2,1}(\mathbb{R})^{\mathcal{X}}$,

$$\ell_{\xi_\pi,2}(\Pi_{\psi,\xi_\pi,\ell_2}\Pi_c\mathcal{T}^\pi\eta,\Pi_{\psi,\xi_\pi,\ell_2}\Pi_c\mathcal{T}^\pi\eta')\leq\gamma^{1/2}\ell_{\xi_\pi,2}(\eta,\eta').$$

Hence the projected distributional Bellman operator has a unique fixed point in $\mathcal{F}_{\psi,S,m}$, by Banach's fixed-point theorem.

Chapter 9: Exact Contraction Chain

For $\eta, \eta' \in \mathcal{M}_{\ell_2, 1}(\mathbb{R})^{\mathcal{X}}$,

$$\begin{aligned} \ell_{\xi_\pi, 2}(\Pi_{\psi, \xi_\pi, \ell_2} \Pi_c \mathcal{T}^\pi \eta, \Pi_{\psi, \xi_\pi, \ell_2} \Pi_c \mathcal{T}^\pi \eta') & \\ & \leq \ell_{\xi_\pi, 2}(\Pi_c \mathcal{T}^\pi \eta, \Pi_c \mathcal{T}^\pi \eta') \\ & \leq \ell_{\xi_\pi, 2}(\mathcal{T}^\pi \eta, \mathcal{T}^\pi \eta') \\ & \leq \gamma^{1/2} \ell_{\xi_\pi, 2}(\eta, \eta'). \end{aligned}$$

From Fixed Features to Learned Features

A deep network replaces the hand crafted features ϕ by a learned representation. Its penultimate layer can still be read as

$$\phi_w(x, a) \in \mathbb{R}^n,$$

followed by linear prediction layer outputting the final things we are interested in.

From DQN to Distributional DQN

Standard DQN predicts one scalar per action:

$$x \mapsto (Q_w(x, a))_{a \in \mathcal{A}}, \quad Q_w(x, a) \approx \mathbb{E}[Z(x, a)].$$

Distributional DQN predicts a return distribution per action:

$$x \mapsto (\eta_w(x, a))_{a \in \mathcal{A}}, \quad \eta_w(x, a) \approx \mathcal{D}(Z(x, a)).$$

For a transition (x, a, r, x') , the target is now a shifted future-return distribution:

$$(b_{r, \gamma})_{\#} \eta_{\tilde{w}}(x', a^*), \quad a^* \in \arg \max_{a' \in \mathcal{A}} Q_{\tilde{w}}(x', a').$$

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Here w are the weights we update. The weights $\tilde{w}$ are a frozen target-network copy, used to have a fixed target.

- **C51**: using the categorical representation.
- **QR-DQN**: using the quantile representation.
- **IQN**: quantile level τ becomes an additional input.

C51: Prediction and Behavior

Using state action categorical representation with fixed $\theta_1 < \dots < \theta_m$, the network outputs $\phi_i(x, a; w)$ and

$$p_i(x, a; w) = \frac{e^{\phi_i(x, a; w)}}{\sum_{j=1}^m e^{\phi_j(x, a; w)}}, \quad \eta_w(x, a) = \sum_{i=1}^m p_i(x, a; w) \delta_{\theta_i}.$$

The induced action value is

$$Q_w(x, a) = \sum_{i=1}^m p_i(x, a; w) \theta_i.$$

Original C51 uses $m = 51$ evenly spaced atoms.

C51 Learning Algorithm

Select the target-network greedy action

$$a_{\tilde{w}}(x') \in \arg \max_{a'} Q_{\tilde{w}}(x', a').$$

Construct and project the target

$$\bar{\eta}(x, a) = \Pi_c(b_{r,\gamma})_{\#} \eta_{\tilde{w}}(x', a_{\tilde{w}}(x')) = \sum_{j=1}^m \bar{p}_j \delta_{\theta_j}.$$

Then minimize the cross entropy loss $L(w) = - \sum_{j=1}^m \bar{p}_j \log p_j(x, a; w)$ and update

$$w \leftarrow w + \alpha \sum_{i=1}^m (\bar{p}_i - p_i(x; w)) \nabla_w \theta_i(x, y; w).$$

This is pretty much the same as the linear CTD update, aside from the need to update vector valued.

QR-DQN: Quantile Prediction

QR-DQN interprets the m outputs as locations of an equally weighted quantile distribution:

$$\eta_w(x, a) = \frac{1}{m} \sum_{i=1}^m \delta_{\theta_i(x, a; w)}, \quad Q_w(x, a) = \frac{1}{m} \sum_{i=1}^m \theta_i(x, a; w).$$

With $a_{\tilde{w}}(x')$ greedy by this mean, the pairwise residuals are

$$\Delta_{ij} = r + \gamma \theta_j(x', a_{\tilde{w}}(x'); \tilde{w}) - \theta_i(x, a; w).$$

Every predicted quantile is compared with every target quantile. We are interested in how much of the target is below/above the quantile.

QR-DQN: Quantile Huber Loss

For $\tau_i = (2i - 1)/(2m)$, QR-DQN minimizes

$$L(w) = \frac{1}{m} \sum_{i,j=1}^m \rho_{\tau_i}^H(\Delta_{ij}), \quad \rho_{\tau}^H(u) = |\mathbb{1}_{\{u < 0\}} - \tau| H_1(u),$$

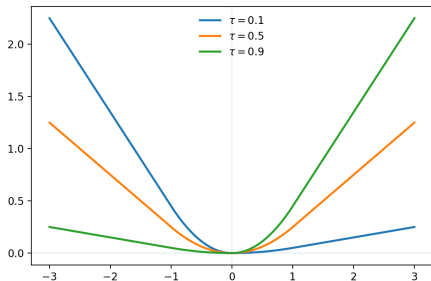
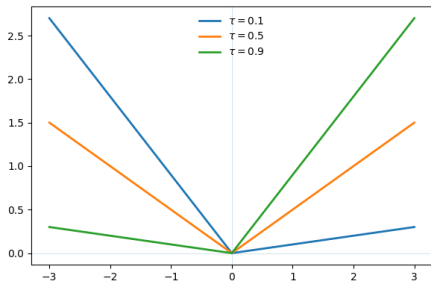
where

$$H_1(u) = \begin{cases} \frac{1}{2}u^2, & |u| \leq 1, \\ |u| - \frac{1}{2}, & |u| > 1. \end{cases}$$

Difference to linear case

- We use the Huber loss to reduce sensitivity to outliers and have a differentiable loss.
- The loss is over all weights simultaneously, not entrywise.

Huber vs Linear Quantile loss



For residual $u = y - \theta_i$, $u > 0$ means the target lies above the predicted quantile, so the prediction is too low. High quantiles therefore give positive residuals a large weight τ_i ; low quantiles give them a small weight. Negative residuals get the opposite weight $1 - \tau_i$.

Particle DQN

QR-DQN represents

$$\eta_w(x, a) = \frac{1}{m} \sum_{i=1}^m \delta_{\theta_i(x, a; w)}$$

where θ_i are quantile levels. MMD-DQN and SinkhornDRL keep the same empirical-measure form

$$\eta_w(x, a) = \frac{1}{m} \sum_{i=1}^m \delta_{z_i(x, a; w)},$$

and treat the z_i as unrestricted deterministic samples.

Overview of Particle DQN

- 1 Define a metric d between return distributions.
- 2 Study whether $\mathcal{T}^\pi$ contracts under $\bar{d}(\eta, \nu) = \sup_{x, a} d(\eta(x, a), \nu(x, a))$.
- 3 Use the empirical version of d as a differentiable DQN loss.

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Overview of Particle DQN

- 1 Define a metric d between return distributions.
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- 3 Use the empirical version of d as a differentiable DQN loss.

The metric **NEEDS** to be differentiable in the particles, this **DOES NOT** hold for the standard Wasserstein.

MMD-DQN: Maximum Mean Discrepancy

Let $\mathcal{H}_k$ be the RKHS of a continuous kernel k and let

$$\psi_\mu = \int k(z, \cdot) d\mu(z) \in \mathcal{H}_k$$

be the kernel mean embedding. The maximum mean discrepancy is

$$\text{MMD}_k(\mu, \nu) = \|\psi_\mu - \psi_\nu\|_{\mathcal{H}_k}.$$

Equivalently, if $Z, Z' \sim \mu$ and $Y, Y' \sim \nu$ independently,

$$\text{MMD}_k^2(\mu, \nu) = \mathbb{E}k(Z, Z') + \mathbb{E}k(Y, Y') - 2\mathbb{E}k(Z, Y).$$

Characteristic kernel

k is characteristic if the embedding $\mu \mapsto \psi_\mu$ is injective:

$$\psi_\mu = \psi_\nu \implies \mu = \nu.$$

For all such k the MMD is a metric.

MMD-DQN: Supremum Metric and Contraction

Nguyen et al. define the distributional-RL metric

$$\text{MMD}_{\infty,k}(\eta, \nu) = \sup_{x,a} \text{MMD}_k(\eta(x, a), \nu(x, a)).$$

Their contraction result requires more than “ k is characteristic.” A kernel must be:

$$k(x+c, y+c) = k(x, y) \quad \text{and} \quad \text{MMD}_k((b_{\gamma,0})_{\#}\mu, (b_{\gamma,0})_{\#}\nu) = \gamma^{\alpha/2} \text{MMD}_k(\mu, \nu).$$

Here $b_{\gamma,0}(z) = \gamma z$ and α is the scale-sensitivity order.

Theorem idea

For mixtures of such kernels, the distributional Bellman operator is

$$\text{MMD}_{\infty,k}(\mathcal{T}^{\pi}\eta, \mathcal{T}^{\pi}\nu) \leq \gamma^{\alpha/2} \text{MMD}_{\infty,k}(\eta, \nu).$$

So $\mathcal{T}^{\pi}$ has the same fixed-point story as in Wasserstein, but only under suitable MMD kernels.

MMD-DQN: Empirics are weird

Theory and empirical observations do not align.

- Unrectified kernels such as

$$k_{\alpha}(x, y) = -\|x - y\|^{\alpha},$$

fulfill the assumption of the Theorem.

- They prove that $\mathcal{T}^{\pi}$ is *not* a contraction in MMD for Gaussian kernels in general.

Yet, in practice the Gaussian Kernel yields better results.

MMD-DQN: Learning with Particles

Algorithm: MMD-DQN particle update

Require: number of particles m , kernel k , discount γ , target weights $\tilde{w}$

Input: sample transition (x, a, r, x')

- 1 Network outputs current particles $Z_w(x, a) = \{z_i(x, a; w)\}_{i=1}^m$, $\eta_w(x, a) = \frac{1}{m} \sum_i \delta_{z_i(x, a; w)}$.
- 2 Select greedy next action

$$a^* \in \arg \max_{a'} \frac{1}{m} \sum_{j=1}^m z_j(x', a'; \tilde{w}).$$

- 3 Construct Bellman target particles

$$y_j = r + \gamma z_j(x', a^*; \tilde{w}), \quad 1 \leq j \leq m.$$

- 4 **Output:** Empirical MMD Bellman loss

$$L_{\text{MMD}}(w) = \frac{1}{m^2} \sum_{i, i'} k(z_i, z_{i'}) + \frac{1}{m^2} \sum_{j, j'} k(y_j, y_{j'}) - \frac{2}{m^2} \sum_{i, j} k(z_i, y_j).$$

SinkhornDRL: Regularized Wasserstein Loss

Sun et al. start from optimal transport. For cost $c(z, y)$,

$$\text{OT}_c(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \int c(z, y) d\pi(z, y).$$

Entropic regularization gives

$$\text{OT}_c^\varepsilon(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \int c d\pi + \varepsilon \text{KL}(\pi \mid \mu \otimes \nu).$$

The sinkhorn divergence is then given by

$$\overline{W}_{c, \varepsilon}(\mu, \nu) = \text{OT}_c^\varepsilon(\mu, \nu) - \frac{1}{2} \text{OT}_c^\varepsilon(\mu, \mu) - \frac{1}{2} \text{OT}_c^\varepsilon(\nu, \nu).$$

Reguralized Wasserstein

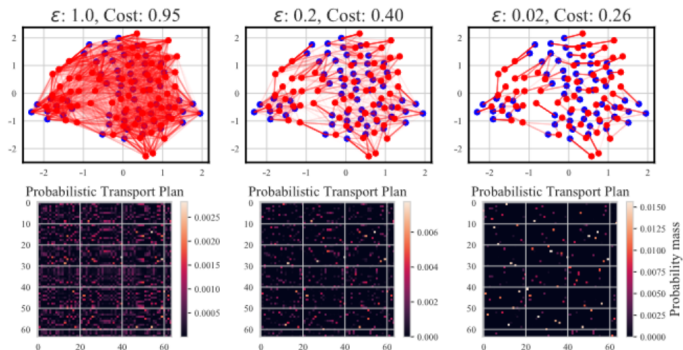


Figure: Comparison of different regularized Wasserstein distances.

SinkhornDRL: Contraction Result

Sun et al. use the standard cost $c(x, y) = \|x - y\|^\alpha$, and define

$$\overline{W}_{c,\varepsilon}^\infty(\eta, \nu) = \sup_{x,a} \overline{W}_{c,\varepsilon}(\eta(x, a), \nu(x, a)).$$

With

$$\lambda_\varepsilon(\mu, \nu) = \frac{\varepsilon \text{KL}(\pi^* \mid \mu \otimes \nu)}{\overline{W}_{c,\varepsilon}(\mu, \nu)} \in (0, 1),$$

the Bellman operator is at least

$$\overline{\Delta}_\varepsilon(\gamma, \alpha) = \gamma^\alpha \left(1 - \sup_{\mu, \nu} \lambda_\varepsilon(\mu, \nu) \right) + \sup_{\mu, \nu} \lambda_\varepsilon(\mu, \nu) \in (\gamma^\alpha, 1)$$

contractive under $\overline{W}_{c,\varepsilon}^\infty$.

SinkhornDRL: Learning with Particles

Algorithm: SinkhornDRL particle update

Require: number of particles m , cost c , regularization ε , discount γ , target weights $\tilde{w}$

Input: sample transition (x, a, r, x')

- 1 Network outputs current particles $Z_w(x, a) = \{z_i(x, a; w)\}_{i=1}^m$, $\eta_w(x, a) = \frac{1}{m} \sum_i \delta_{z_i(x, a; w)}$.
- 2 Select greedy next action

$$a^* \in \arg \max_{a'} \frac{1}{m} \sum_{j=1}^m z_j(x', a'; \tilde{w}).$$

- 3 Construct Bellman target particles

$$y_j = r + \gamma z_j(x', a^*; \tilde{w}), \quad 1 \leq j \leq m.$$

- 4 **Output:** approximate Wasserstein using Sinkhorn:

$$L_{\text{RW}}(w) = \overline{W}_{c, \varepsilon} \left(\frac{1}{m} \sum_i \delta_{z_i(x, a; w)}, \frac{1}{m} \sum_j \delta_{y_j} \right).$$

SinkhornDRL: Interpolating Wasserstein and MMD

The regularization parameter moves between Wasserstein and MMD:

$$\varepsilon \rightarrow 0 \quad \Rightarrow \quad \overline{W}_{c,\varepsilon}(\mu, \nu) \rightarrow 2W_c^\alpha(\mu, \nu),$$

$$\varepsilon \rightarrow +\infty \quad \Rightarrow \quad \overline{W}_{c,\varepsilon}(\mu, \nu) \rightarrow \text{MMD}_{k_\alpha}^2(\mu, \nu), \quad k_\alpha = -c.$$

Algorithm	Distribution divergence	Representation Z_θ	Rate of $\mathcal{T}^\pi$
C51	Cramer distance	Categorical distribution	$\sqrt{\gamma}$
QR-DQN-1	Wasserstein distance	Quantiles	γ
MMD-DQN	MMD	Samples	$\gamma^{\alpha/2} (k_\alpha)$
SinkhornDRL	Sinkhorn divergence	Samples	$\gamma (\varepsilon \rightarrow 0), \gamma^{\alpha/2} (\varepsilon \rightarrow +\infty)$

SinkhornDRL: Atari Comparison

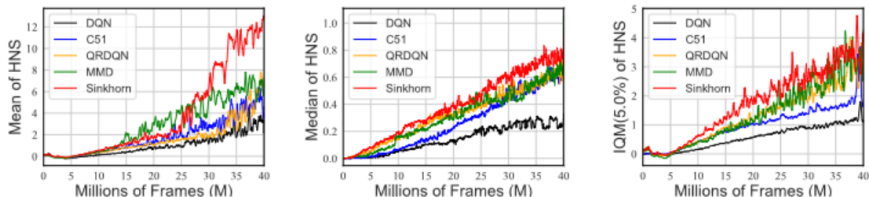


Figure: Comparison of various algorithms on 55 Atari games.

References

- Bellemare, Dabney, Rowland (2023): *Distributional Reinforcement Learning*
- Nguyen et al. (2020): Distributional Reinforcement Learning via Moment Matching.
- Sun et al. (2022): Distributional Reinforcement Learning with Regularized Wasserstein Loss.